

School Management System Requirements Spec

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School Management System — Requirements Spec

Grade 1-3 Integrated School | Kenya Scope: Attendance, Academic Reports, Fees Payment

1. User Roles

Role	Access
Admin	Full system access — manages teachers, classes, students, invitations, academic report approval/sending, fees, public website content
Teacher	Access limited to classes assigned to them
Parent/Guardian	View-only — not an active system user in the operational sense (no data entry, no actions taken in-system). Can log in to view their child's attendance, academic report, and fee statement. All notifications (absence, report-ready, fee reminders, invitations) are pushed to them via SMS regardless of whether they log in.

Note on “viewer” role: Since parents don't take actions in the system (they don't submit anything), their portal can be kept intentionally simple — a read-only dashboard rather than a full account with settings/preferences. SMS remains the primary channel; the portal login is a secondary, optional way to check details.

2. Teacher Flows

2.1 Attendance (Daily Register)

1. Teacher logs in → sees **list of classes assigned to them** (a teacher may have more than one class)
2. Teacher opens a class → sees **full roster of registered students** in that class
3. Teacher marks each student **Present / Absent** for the day
4. Before submitting, teacher sees a **review screen: list of students marked Absent**
5. On the review screen, teacher can **check a box next to each absent student** to trigger an SMS to that student's parent/guardian
6. Teacher clicks **Submit Register**
7. On submit:

- Attendance is saved and **locked** (edits after this require admin override)
- SMS is sent **only now**, only to the parents of students whose “send SMS” box was checked
- SMS content: student did not report to school today

Data needed: class list per teacher, student roster per class, attendance status per student per day, submit timestamp, SMS-sent flag per record.

Assumption: A teacher can un-check the SMS box for a specific absent student — e.g., if the parent already called in to explain the absence, no need to notify them. Confirm if this matches intent.

2.2 Academic Reports

1. Teacher opens a student (or a class, generating one per student) → fills in a **performance report**
2. Report is saved as **draft** under that student
3. Report becomes visible to parents only when **admin sends it** (see Admin flow 3.3) — teacher does not send directly to parents

Open question: What should the report contain? Typical structure for Grade 1–3 (narrative, not letter grades): - Subject-by-subject rating (e.g., Excellent / Good / Needs Improvement) per subject - Teacher’s general comments - Attendance summary for the term (can auto-pull from Module 1) - Term and academic year

Let me know if the school already has a report card format/template — that will shape the input form design.

3. Admin Flows

3.1 Assign Teachers to Classes

- Admin creates/edits a class (grade + section)
- Admin assigns one (or more, e.g. class teacher + subject teacher) teacher(s) to that class

3.2 Admit Student

- Admin registers a new student: name, grade/class, date of birth, parent/guardian name(s) + phone number(s)
- Student appears immediately on their assigned teacher’s roster

3.3 Send Academic Reports

- Admin reviews teacher-submitted draft reports
- Admin sends report → parent is notified (SMS with a link, or SMS + report visible in parent portal)

Assumption: Admin reviews before sending, rather than reports going out automatically. This gives a quality-control step, but adds manual work — confirm this is wanted, versus teacher-submit = auto-send.

3.4 Send Invitations / Messages to Parents

- Admin can send a message to:
 - A single parent
 - All parents in a class
 - All parents in a grade
 - All parents school-wide
- Used for: event invitations, general announcements, closures, reminders
- Delivered via SMS (and optionally shown in parent portal)

3.5 Fee Statements & Reminders

- Admin can view **any student’s fee statement**: invoices issued, amounts paid, balance, payment dates/references
- Admin can view a **school-wide fees dashboard**: filterable by grade/class, sortable by balance owed, paid/pending/overdue status

- Admin can select one student, several, or a whole class/grade with an outstanding balance → **send a fee balance reminder SMS**
- Reminder SMS includes: student name, amount owed, due date (optionally a payment link/USSD prompt)

3.6 Public Website Content (Calendar & Announcements)

- Admin can **post calendar events** (e.g., term dates, holidays, sports day, parent-teacher meetings) visible on the public website
 - Admin can **post general information/announcements** to the public website (separate from the SMS-based parent messaging in 3.4 — this is public-facing, not targeted)
 - No developer involvement needed for these two content types day-to-day; everything else on the public site (About, Admissions, Contact) remains static and developer-maintained, per earlier discussion
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4. Parent Flow (View-Only)

- Parent logs in → sees **all their children** (a parent may have multiple kids in the school, each possibly in a different class)
 - No data entry or actions — this is a read-only dashboard. Per child, parent can view:
 - Attendance history
 - Academic report (once sent by admin)
 - Fee balance and payment history (see Module 5)
 - Parent receives SMS for:
 - Absence notification (same day, post-register-submission)
 - New academic report available
 - General announcements/invitations from admin
 - Fee reminders/receipts
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5. Fees Payment Module

(Carried over from earlier discussion — confirm this is still in scope alongside attendance/reports)

- Admin sets fee structure per grade/term
 - System generates an invoice per student per term
 - Parent pays via **M-Pesa STK Push** (through Pesapal or IntaSend)
 - Payment auto-updates balance, triggers SMS receipt
 - Admin dashboard: paid / pending / overdue, filterable by class/grade
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6. Notifications — Shared Layer

All SMS (absence alerts, report-ready alerts, invitations, fee reminders/receipts) run through one shared service: - **Africa's Talking** (SMS gateway) — single integration point used by all modules - Every SMS logged (sent to, content, timestamp, related record) for audit/traceability

7. Core Data Model (minimal)

- **User** (id, role: admin/teacher/parent, name, phone, email, password)
- **Class** (id, grade, section, assigned_teacher_id)
- **Student** (id, name, dob, class_id, parent_id[s])
- **AttendanceRecord** (id, student_id, date, status, marked_by, submitted_at)
- **AcademicReport** (id, student_id, term, subject_ratings, comments, status: draft/sent, created_by, sent_by,

sent_at)

- **Invoice** (id, student_id, term, amount_due, amount_paid, status)
 - **Payment** (id, invoice_id, amount, mpesa_ref, timestamp)
 - **Message** (id, sender_id, target_scope: student/class/grade/school, content, sent_at)
 - **SMSLog** (id, recipient_phone, content, related_record_type, related_record_id, status, timestamp)
 - **CalendarEvent** (id, title, description, date, created_by, is_public)
 - **PublicPost** (id, title, content, posted_by, posted_at, is_public)
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8. Deliberately Out of Scope (for this phase)

- Homework/assignment posting
 - Gradebook (numeric grading) — replaced by simplified narrative report
 - Public marketing website
 - Teacher-to-parent direct chat/messaging threads
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9. Open Questions for the School

1. Can a teacher un-check specific absent students before sending SMS (e.g., parent already informed)?
2. Should academic reports require admin approval before reaching parents, or send automatically once a teacher submits?
3. What should the academic report template actually contain — does the school have an existing report card format?
4. Should a class have exactly one teacher, or can multiple teachers (e.g., subject specialists) be attached to one class?